

Kidney Allocation Under Observational Learning

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Abstract

The kidney allocation system suffers from high demand and much lower supply, with the US National Kidney waiting list over 90,000 people long. Deceased-donor kidneys are perishable, and an organ must be placed within a short viability window or it is discarded. Today, the United States discards roughly 22% of procured kidneys, kidneys which could save a life among the 30% of waiting patients who die or are removed from the list before receiving a kidney. In this paper, we study why a sequential offer mechanism with a search budget results in wasted organs. We also examine which alternative mechanisms recover the loss. We embed a single behavioral model, grounded in a private reservation threshold followed by a logistic accept/decline that depends on queue position and on the number of prior declines observed for the offered organ to model both desperation and observational learning, inside four allocation mechanisms: *omniscient* (first-best), *sequential* (the existing mechanism), *batching*, and a family of *threshold* (pre-declaration) mechanisms indexed by declaration granularity K . The decline-count term represents observational learning, the cascading decline effect documented by Zhang (2010). We index the strength of this effect with a parameter δ and sweep over $\{0, 0.3, 1.0\}$. Simulating each mechanism against a common calibrated environment, we find: (i) a continuous threshold mechanism attains near-first-best discard rates at every δ , neutralizing the cascading effect; (ii) under common random numbers it *strictly Pareto-dominates* batching and sequential at the individual-patient level for $\delta > 0$ (no patient is worse off, many are better off); (iii) the gains concentrate almost entirely on marginal (high-KDPI) organs, while premium organs are placed at the front of the queue under every mechanism; (iv) the cost of coarsening the declaration grows with δ , but *cutoff placement dominates bucket count* (the current OPTN binary opt-in at KDPI=85 is near-optimal among binary cuts); and (v) threshold and batching are complements (a hybrid strictly improves on either alone). All conclusions are conditional on the observational-learning interpretation of decline cascades. We also discuss the competing private-information interpretation of Agarwal et al. (2021) and the empirical test we could run in further research.

1 Introduction

When someone dies and becomes an organ donor, their kidneys must be transplanted quickly. A kidney removed from a deceased donor stays viable for transplant about roughly 12 to 24 hours before the tissue degrades enough that surgeons will no longer use it. Among kidneys, the target is usually 18 hours between retrieval and transplant. After that, the kidney is discarded, wasted, even though tens of thousands of patients are waiting. The U.S. discards roughly 22% of procured deceased-donor kidneys, a figure that has risen rather than fallen over the past decade (Mohan et al., 2023). The French system, with similar donor characteristics, discards under 10% of donated kidneys.

The current U.S. allocation system works roughly as follows. When a kidney becomes available, the procurement organization consults a ranked list of waiting patients based on different factors such as waiting time, urgency, and compatibility. The patient at the top is offered the kidney first,

and their transplant center has up to one hour to accept or decline. If they decline, the offer moves to the next patient, and so on down the list. Each “no” eats into the kidney’s viability clock. If a dozen patients in a row decline, the organ has exhausted its window and is discarded. The organ is discard even though the thirteenth, or the twentieth, or the hundredth candidate might have said yes.

This is one of the cleanest instances in market design of a fundamental tension: **the mechanism that matches goods to people operates under a budget, and the budget is consumed by declines**. Husain et al. (2019) report that the median recipient of a deceased-donor kidney received 17 prior offers (made on behalf of higher-priority patients) before the organ reached them. However, 30% of waitlisted patients die or are removed from the list before ever receiving a transplant. The organ at the top of the match run cannot wait for the right answer. Instead, it must find an answer within its budget before time runs out.

This paper compares four mechanisms which attempts to increase the efficiency of this problem and decrease the rate of organ discards. We specify a single model of the allocation environment and of patient behavior, then evaluate each mechanism inside that common environment. The question is which mechanism produces the most transplants, with well-matched recipients, without unfairly disadvantaging high-priority patients waiting in line. We treat the problem in the language of market design: a mechanism is a map from the state of the waiting list and an arriving object to an allocation, and we ask which maps are efficient, which are Pareto-comparable, and how their performance depends on the strength of the informational externality that declines impose on later patients.

2 Methods

Each mechanism is a different policy for finding a recipient for an available kidney.

Omniscient. This is a first-best benchmark. We assume that the mechanism knows every patient’s preferences and instantly identifies the highest-priority patient who would accept the organ, allocating it with no further friction. This is unattainable in practice since real designers neither observe preferences nor act instantaneously, but this method bounds what any other mechanism can achieve.

Sequential. The current mechanism in the United States. The organ is offered to one patient per period, walking down the priority queue one-at-a-time, until someone accepts or the budget of H periods is exhausted. This approximates current OPTN allocation, abstracting from parallel offering to multiple centers and geographic priority bands.

Batching. A proposed reform to the current system. Instead of one offer per period, the mechanism offers simultaneously to the next m patients in priority order. Whoever among them accepts receives the organ. Ties are broken in favor of the highest-ranked acceptor. If all m decline, the next batch is offered the following period. For example, with $m = 5$ and $H = 12$, the mechanism can reach up to 60 candidates rather than 12. OPTN pilots of batched offers (batches of four to five) report reduced workload and improved utilization in practice, and we quantify that improvement here.

Threshold. A pre-declaration redesign based on Su and Zenios (2005). Each patient declares, in advance, which organ-quality ranges they would accept. When an organ arrives, the mechanism offers it only to patients whose declaration covers the organ’s quality, in priority order. Non-declarers are skipped at no budgetary cost. We study five granularities of the declaration question: a binary opt-in (based on the current OPTN cut-off for high-KDPI organs), three buckets, five, ten, and continuous self-declaration (the Su-Zenios theoretical ideal).

To compare these fairly, patient behavior must be invariant to the mechanism: the mechanism does not change what a patient wants, only *how the system discovers it and how often it asks*. We therefore specify one behavioral model and run all four mechanisms against the same realized environment.

3 Model

We describe the environment, the preference primitive, and the behavioral acceptance rule. As previously mentioned, the environment and behavior are held fixed across mechanisms, and only the mechanism’s offer policy varies.

3.1 Objects (Kidneys)

A kidney has quality $\theta \in [0, 1]$, with $\theta = 1$ the best attainable organ and $\theta = 0$ the worst. We identify θ with a monotone transform of the existing OPTN Kidney Donor Profile Index (KDPI), following the equation $\theta = 1 - \text{KDPI}/100$. For example, a “marginal” organ at the OPTN special-consent boundary KDPI=85 has $\theta = 0.15$. A premium organ with KDPI=15 has $\theta = 0.85$. Because KDPI is a percentile score by construction, the population distribution of quality is uniform:

$$\theta_k \stackrel{\text{iid}}{\sim} \text{Uniform}[0, 1], \quad k = 1, 2, 3, \dots \quad (1)$$

Organs arrive one at a time, indexed by k , each with a viability budget of H discrete offer periods (“hours”) before discard.

3.2 Agents (Patients) and Preferences

Each patient i has a private **reservation threshold** $\tilde{\theta}_i \in [0, 1]$ and is willing to accept organ with quality θ if and only if the quality meets their personal threshold:

$$\text{patient } i \text{ would accept } \theta \iff \tilde{\theta}_i \leq \theta. \quad (2)$$

This is the patient’s *preference*, which may be distinct from *behavior* (specified below). For example, a patient with $\tilde{\theta}_i = 0.05$ has weak outside options and accepts nearly any organ. However, one with $\tilde{\theta}_i = 0.95$ is extremely selective. Thresholds are drawn

$$\tilde{\theta}_i \stackrel{\text{iid}}{\sim} \text{Beta}(2, 5), \quad \mathbb{E}[\tilde{\theta}_i] = \frac{2}{7} \approx 0.286, \quad (3)$$

a right-skewed distribution under which most patients accept moderate organs but only a minority accept the most marginal ones. The calibration matches the roughly 30% of OPTN waitlist patients who opt in to receive high-KDPI organs, such that $\mathbb{P}(\tilde{\theta}_i \leq 0.15) = \int_0^{0.15} f_{\text{Beta}(2,5)}(t) dt \approx 0.30$.

The threshold $\tilde{\theta}_i$ is **private information**, so we assume that the mechanism cannot observe it directly. This is the informational friction at the center of the design problem. The four mechanisms differ precisely in how they cope with the unobservability of $\tilde{\theta}_i$. Omniscient sees it by assumption, sequential and batching never elicit it, and the threshold family elicits a coarsened report of it in advance. (We note that the continuous threshold method is unattainable in practice, as in the real world patients likely do not know their exact threshold.)

Patients arrive at rate ρ per organ, so the queue grows over time. Priority is determined by arrival order (earlier arrivals rank higher), a version of the OPTN points system that we treat as pure time-on-list. Formally, immediately after organ k is processed, $A_k \sim \text{Poisson}(\rho)$ new patients join the tail of the queue, each with an independent threshold drawn from (3). We set $\rho = 1.6$, matching the U.S. ratio of roughly 40,000 annual waitlist additions to 25,000 deceased-donor transplants.

3.3 Behavioral Acceptance Model

Consider patient i offered organ θ . Let n denote i 's position in the queue (measured from the front, so $n = 1$ is the highest-priority waiting patient) and let k denote the number of declines already recorded for *this* organ at the moment i is asked. We use a two-stage acceptance rule: a deterministic *threshold gate* followed by a stochastic *logistic* response.

Definition 1 (Acceptance function). Let $\sigma(x) = (1 + e^{-x})^{-1}$ be the standard logistic function. The probability that patient i accepts organ θ when at position n having observed k prior declines is

$$\pi(\theta, \tilde{\theta}_i, n, k) = \begin{cases} 0, & \tilde{\theta}_i > \theta \quad (\text{threshold gate}) \\ \sigma(\alpha + \beta(n-1) - \delta k), & \tilde{\theta}_i \leq \theta \quad (\text{logistic}). \end{cases} \quad (4)$$

Conditional on the offer, i accepts iff an independent draw $U_i \sim \text{Uniform}[0, 1]$ satisfies $U_i < \pi(\theta, \tilde{\theta}_i, n, k)$.

The three structural parameters are:

- $\alpha = \text{logit}(0.9) = \ln 9 \approx 2.197$, the baseline log-odds. At the front of the queue with no prior declines ($n = 1, k = 0$), a threshold-eligible patient accepts with probability $\sigma(\alpha) = 0.9$. The 10% residual decline captures non-preference frictions such as surgical scheduling conflicts, last-minute donor concerns, and family decisions.
- $\beta = 0.10$, the **position (desperation) effect**. Deeper-queued patients accept more readily even conditional on eligibility, reflecting their longer waiting time until they receive their next offer and worse outside options. Each step deeper raises the acceptance log-odds by β .
- $\delta \geq 0$, the **cascading rejections strength**. Each prior decline lowers the next patient's acceptance log-odds by δ , modeling how concerns about a particular organ could increase with the number of rejections. We sweep $\delta \in \{0, 0.3, 1.0\}$. At $\delta = 0$, prior declines are uninformative. At $\delta = 1.0$, each decline divides the acceptance odds by $e \approx 2.72$.

Because (4) is linear in the log-odds, it is equivalently multiplicative in the odds. Writing $o = \pi/(1 - \pi)$ for the acceptance odds of an eligible patient,

$$o(\theta, \tilde{\theta}_i, n, k) = e^\alpha \cdot (e^\beta)^{n-1} \cdot (e^{-\delta})^k, \quad (5)$$

so each additional queue step multiplies the odds by $e^\beta \approx 1.105$ and each prior decline multiplies them by $e^{-\delta}$. The cascade is therefore a geometric decay in the odds, and the linear-in- k specification is exactly the posterior that a Bayesian patient would hold if each decline were a conditionally independent signal of equal strength about a latent organ defect (the log-posterior-odds add one log-likelihood-ratio per signal). We return to this in §13.

A subtle but important feature is that the counter k does not distinguish a threshold-gate auto-decline ($\tilde{\theta}_i > \theta$) from a logistic decline ($U_i \geq \sigma(\cdot)$). We model it this way because a downstream patient observes only that the previous candidate said “no,” not why the kidney was rejected by previous patients. Both decline types therefore increment k and contribute to the cascading declines signal, matching the observational setup of Zhang (2010), in which the econometrician (like the patient) sees declines but not their reasons.

3.4 Behavioral Modeling Choices

We explain four modeling choices below:

The threshold gate. In a frictionless world, a patient accepts any organ above their reservation utility and declines any below. The gate encodes exactly this and guarantees *ex post* rationality, given that patients are given a choice of whether or not to accept. No patient is ever assigned an organ they did not want (formally, every recipient i of organ θ satisfies $\tilde{\theta}_i \leq \theta$, a property we verify in §11). Without the gate the model would predict patients accepting organs they would regret, which the structural literature finds rare in equilibrium.

The logistic layer. A pure threshold rule predicts 100% acceptance among eligible patients, which is empirically false—OPTN decline rates are well above zero even among patients who pre-opted in to marginal organs. The 10% baseline captures the universe of non-preference reasons an offer can fail, consistent with the 10–20% decline rates among high-acceptance patients reported by Husain et al. (2019).

The position effect. Queue depth correlates with waiting time, medical urgency, and exhausted outside options, all of which raise acceptance. A front-of-queue patient retains the option to wait for a better organ. However, a deep patient may not due to the extended waiting time until the next offer. The term $\beta(n - 1)$ is the simplest monotone encoding of this desperation (though future experiments should change this modeling).

The cascading demands effect. This is the most contested feature. Zhang (2010) provides direct evidence of observational learning in U.S. kidney offer responses: candidates lower on the list decline marginal organs more often than observables predict. Agarwal et al. (2021) offer a competing interpretation in which declines reflect *privately observed organ characteristics*, so that later patients are not learning from predecessors but independently observing the same defects. The two interpretations generate similar static decline patterns but *opposite* comparative statics with respect to mechanism design: if declines are social learning, a mechanism that hides or freezes the decline signal helps; if declines are private information, it does not. Our model takes Zhang’s interpretation at face value, which gives mechanisms the strongest lever and makes the comparison sharp. **We note that all results should be read as conditional on $\delta > 0$ being the right description of the world**; the $\delta = 0$ column is exactly the Agarwal-world benchmark, and we report it throughout to show the effect across both worlds.

Remark 1 (One behavioral model, four mechanisms). *The function π in (4) is identical across all mechanisms except omniscient. What varies is which (n, k) pairs are fed into π and which patients are offered the organ at all. A patient’s response to a given offer is defined by their position, the observed decline count, and their private threshold. This response is invariant to the mechanism. This is what makes the comparison a clean test of mechanism design rather than of behavioral assumptions.*

4 Formal Mechanisms

We now define each mechanism precisely as a map on the waiting list.

Common Notation

- $\mathcal{Q} = (p_1, p_2, \dots, p_{|\mathcal{Q}|})$ is the queue when organ θ arrives, ordered by priority (p_1 at the front).
- $\tilde{\theta}_i \in [0, 1]$ is p_i ’s private threshold; p_i would accept θ iff $\tilde{\theta}_i \leq \theta$.

- $H = 12$ is the viability budget in offer periods. We choose 12 hours to model time required to transport and transplant kidneys, rather than the full 18 hour budget from retrieval until transplant.
- $U_i \sim \text{Uniform}[0, 1]$ is p_i 's acceptance draw, independent across offers.
- The acceptance function $\pi(\theta, \tilde{\theta}_i, n, k)$ of (4) governs all mechanisms except omniscient.

Definition 2 (Mechanism). A mechanism is a (possibly randomized) map $M : (\mathcal{Q}, \theta) \mapsto \mathcal{Q}'$ returning either $\mathcal{Q} \setminus \{p_{i^*}\}$ for some recipient i^* (allocation) or $\mathcal{Q}$ unchanged (discard). The organ is *discarded* iff $M(\mathcal{Q}, \theta) = \mathcal{Q}$.

For a stream of organs $\theta_1, \dots, \theta_N$ with interleaved arrivals, applying M organ-by-organ induces the two primary outcome statistics, the **discard rate** $D(M)$ and the **mean placement** $P(M)$ (average queue rank of recipients), defined formally in §5.

4.1 Omniscient (first-best benchmark)

Omniscient is the only mechanism that does not use π . Instead, it assumes preferences are observable and decisions deterministic:

$$M_{\text{Omn}}(\mathcal{Q}, \theta) = \begin{cases} \mathcal{Q} \setminus \{p_{i^*}\}, & i^* = \min\{i : \tilde{\theta}_i \leq \theta\}, \\ \mathcal{Q}, & \{i : \tilde{\theta}_i \leq \theta\} = \emptyset \text{ (discard)}. \end{cases} \quad (6)$$

There is no budget and no noise in this model since we assume fully observable characteristics. Our search depth is unbounded, and eligible patients accept with certainty.

Proposition 1 (First-best discard floor). *For any realized queue $\mathcal{Q}$ and organ θ , omniscient discards θ iff no waiting patient is willing to accept it: $\tilde{\theta}_i > \theta$ for all $p_i \in \mathcal{Q}$. Consequently $D(M) \geq D(M_{\text{Omn}})$ for every mechanism M acting on the same realized environment, and any excess $D(M) - D(M_{\text{Omn}})$ is attributable to information frictions such as a finite search budget, logistic noise, or the declines cascade.*

Proof. A discard under M_{Omn} occurs only when the eligible set is empty, by (6); in that event every mechanism discards, since no offer can be accepted (the gate forces $\pi = 0$ for all patients). Conversely, whenever the eligible set is nonempty, M_{Omn} allocates with certainty, so its discard rate is a pointwise lower bound. Aggregating over the organ stream preserves the inequality. $\square$

Omniscient's mean placement is high since it reaches as deep as needed. Simply, this mechanism has high mean placement precisely because it never wastes an offer on a non-acceptor.

4.2 Sequential ($m = 1$)

At period h the organ is offered to p_h (queue position $n = h$). Every prior period produced a decline, so the previous-discard counter is exactly $k = h - 1$. The patient accepts iff

$$U_h < \pi(\theta, \tilde{\theta}_h, n=h, k=h-1) = \mathbf{1}[\tilde{\theta}_h \leq \theta] \cdot \sigma(\alpha + (\beta - \delta)(h - 1)). \quad (7)$$

Algorithm 1: Sequential mechanism

Input: queue $\mathcal{Q}$, organ quality θ , cascade strength δ

```

1 for  $h \leftarrow 1$  to  $\min(H, |\mathcal{Q}|)$  do
2   sample  $U_h \sim \text{Uniform}[0, 1]$ 
3   if  $\tilde{\theta}_h \leq \theta$  and  $U_h < \sigma(\alpha + (\beta - \delta)(h - 1))$  then
4     return  $\mathcal{Q} \setminus \{p_h\}$                                      //  $p_h$  receives the organ
5 return  $\mathcal{Q}$                                                          // discard

```

Proposition 2 (Cascade dominance in sequential). *In the sequential mechanism the position and cascading discards effects collapse into a single coefficient $(\beta - \delta)$ on depth: the eligible-patient acceptance probability at period h is $\sigma(\alpha + (\beta - \delta)(h - 1))$. Hence acceptance is increasing in depth iff $\delta < \beta$ and decreasing iff $\delta > \beta$. For all $\delta > 0.10$ the cascade dominates desperation and the per-offer acceptance probability decays geometrically in the odds at rate $e^{-(\delta - \beta)}$.*

Proof. Immediate from (7): in sequential, $n = h$ and $k = h - 1$ are perfectly collinear, so $\beta(n - 1) - \delta k = (\beta - \delta)(h - 1)$. The monotonicity and decay-rate claims follow from the sign of $\beta - \delta$ and (5). $\square$

Proposition 2 isolates why sequential is fragile. It offers to every patient in line regardless of eligibility, so each unavoidable decline both consumes budget and amplifies the cascade, and at empirically relevant δ the two effects compound.

4.3 Batching (m offers per period)

Define the period- h batch $B_h = \{p_{(h-1)m+1}, \dots, p_{hm}\} \cap \mathcal{Q}$. All members of B_h observe the *same* frozen previous-decline count $k_h = (h - 1)m$. Each $p_i \in B_h$ at position n_i independently accepts iff

$$U_i < \pi(\theta, \tilde{\theta}_i, n_i, k_h) = \mathbf{1}[\tilde{\theta}_i \leq \theta] \cdot \sigma(\alpha + \beta(n_i - 1) - \delta(h - 1)m). \quad (8)$$

Let $A_h = \{i \in B_h : \text{accept}_i\}$. If $A_h \neq \emptyset$ the organ goes to the highest-priority acceptor $i^* = \min A_h$; otherwise the mechanism proceeds to B_{h+1} .

Algorithm 2: Batching mechanism with batch size m

Input: queue $\mathcal{Q}$, organ quality θ , batch size m , cascade strength δ

```

1 for  $h \leftarrow 1$  to  $H$  do
2    $B_h \leftarrow \{p_{(h-1)m+1}, \dots, p_{hm}\} \cap \mathcal{Q}$ 
3   if  $B_h = \emptyset$  then return  $\mathcal{Q}$ 
4    $k_h \leftarrow (h - 1)m$ ;  $A_h \leftarrow \emptyset$ 
5   foreach  $p_i \in B_h$  at position  $n_i$  do
6     sample  $U_i \sim \text{Uniform}[0, 1]$ 
7     if  $\tilde{\theta}_i \leq \theta$  and  $U_i < \sigma(\alpha + \beta(n_i - 1) - \delta k_h)$  then  $A_h \leftarrow A_h \cup \{i\}$ 
8   if  $A_h \neq \emptyset$  then return  $\mathcal{Q} \setminus \{p_{\min A_h}\}$ 
9 return  $\mathcal{Q}$                                                          // discard

```

The structural difference from sequential is the cascading decline counter. In sequential k increments by one each step ($k_h = h - 1$); in batching it increments in jumps of m ($k_h = (h - 1)m$) and is *frozen within a batch*, so a patient never observes a batchmate's decline. This information-freezing blunts the cascade. With $m = 5$ and $H = 12$, batching reaches up to $Hm = 60$ candidates versus sequential's 12.

Proposition 3 (Batching weakly improves on sequential). *Under common random draws, for any organ and queue, the set of acceptors that batching ($m \geq 1$) can reach within budget contains the singleton sequential reaches, and the within-batch declines count $k_h = (h - 1)m$ is weakly below the sequential count $k = h - 1$ at the same patient for the patients reached early. Both the larger candidate pool (Hm vs. H) and the frozen, weakly-smaller declines count weakly raise the allocation probability, so $D(M_{\text{Bat}}) \leq D(M_{\text{Seq}})$ in expectation.*

We verify Proposition 3 numerically in Sections 6 and 9, and quantify the batch-size margin there.

4.4 Threshold (K buckets)

Fix cutoffs $0 = c_0 < c_1 < \dots < c_{K-1} < c_K = 1$ partitioning $[0, 1]$ into buckets $\mathcal{B}_b = [c_b, c_{b+1})$, $b = 0, \dots, K - 1$. Each patient declares the lowest-quality bucket they accept,

$$b_i = \max\{b \in \{0, \dots, K - 1\} : c_b \leq \tilde{\theta}_i\}. \quad (9)$$

Proposition 4 (Truthful declaration is dominant). *If patients know their own thresholds and the mechanism offers eligible patients in priority order, the declaration (9) is (weakly) dominant-strategy incentive-compatible: no misreport raises a patient’s probability of receiving an acceptable organ. Over-declaring (claiming a lower bucket) exposes the patient to offers of organs below θ_i , which the gate forces them to decline—wasting their turn without benefit; under-declaring forgoes acceptable offers. This is the Su and Zenios (2005) result specialized to our environment; we assume truthful declaration throughout.*

Eligibility and allocation. For organ θ define its bucket $b_\theta = \max\{b : c_b \leq \theta\}$. Patient p_i is eligible iff $b_i \leq b_\theta$. The eligible set in priority order is

$$\mathcal{E}_\theta(\mathcal{Q}) = \langle p_{i_1}, p_{i_2}, \dots \rangle, \quad i_1 < i_2 < \dots, \quad b_{i_j} \leq b_\theta. \quad (10)$$

The mechanism makes at most H offers to eligible patients, maintaining a declines counter k that starts at 0 and increments on every decline (gate or logistic). Crucially, the desperation term uses each patient’s queue position n_i , not their rank within $\mathcal{E}_\theta$:

$$\text{accept}_i = \mathbf{1}[\tilde{\theta}_i \leq \theta] \cdot \mathbf{1}[U_i < \sigma(\alpha + \beta(n_i - 1) - \delta k)]. \quad (11)$$

Algorithm 3: Threshold mechanism with K buckets

Input: queue $\mathcal{Q}$, organ quality θ , cutoffs $\{c_b\}$, cascade strength δ

```
1  $b_\theta \leftarrow \max\{b : c_b \leq \theta\}; \quad k \leftarrow 0; \quad j \leftarrow 0$ 
2 foreach  $p_i \in \mathcal{Q}$  in priority order do
3   if  $b_i > b_\theta$  then continue                                     // ineligible: skip at no cost
4    $j \leftarrow j + 1$ 
5   if  $j > H$  then break
6   sample  $U_i \sim \text{Uniform}[0, 1]$ 
7   if  $\tilde{\theta}_i > \theta$  then
8      $k \leftarrow k + 1$                                            // within-bucket mismatch: auto-decline
9   else if  $U_i < \sigma(\alpha + \beta(n_i - 1) - \delta k)$  then
10    return  $\mathcal{Q} \setminus \{p_i\}$ 
11  else
12     $k \leftarrow k + 1$                                            // logistic decline
13 return  $\mathcal{Q}$                                                     // discard
```

Two structural features distinguish threshold from sequential and batching:

1. **The eligibility filter.** Ineligible patients consume zero budget, since this mechanism can already weed out those who have not selected relevant buckets. Sequential and batching offer to whoever is next regardless of threshold, wasting budget and generating cascade signal on patients who will decline.
2. **Position, not eligible-rank, drives desperation.** The term $\beta(n_i - 1)$ uses the true queue position, encoding that desperation accrues from time-on-list, not from the offering process.

The five variants. We test five declaration granularities:

- $K = 2$ (**OPTN**): single cutoff $c_1 = 0.15$ (KDPI=85). This is the current OPTN binary opt-in on the waiting list questionnaire.
- $K = 3$: uniform cutoffs $\{1/3, 2/3\}$.
- $K = 5$: uniform cutoffs $\{0.2, 0.4, 0.6, 0.8\}$.
- $K = 10$: uniform cutoffs $\{0.1, 0.2, \dots, 0.9\}$.
- $K = \infty$ (**continuous**): the patient declares $\tilde{\theta}_i$ exactly; eligibility is $\tilde{\theta}_i \leq \theta$. This is the Su-Zenios ideal.

Proposition 5 (Within-bucket cascades under coarse K). *For finite K , a patient with $\tilde{\theta}_i \in (c_{b_i}, \theta]$ is eligible for organ θ (their bucket covers it) yet auto-declines (their precise threshold rejects it), since $\tilde{\theta}_i > \theta$ is possible within $\mathcal{B}_{b_i}$ when θ lies in the same bucket. Each such within-bucket auto-decline increments k and feeds the cascade. As $K \rightarrow \infty$ the bucket widths vanish and within-bucket auto-declines disappear: the only remaining source of decline is the logistic noise term in (11).*

Proof. Eligibility requires $b_i \leq b_\theta$; for a patient in the kidney's own bucket ($b_i = b_\theta$) eligibility holds while $\tilde{\theta}_i$ may exceed θ within $[c_{b_\theta}, c_{b_\theta+1})$, triggering the gate. The measure of such patients is bounded by the bucket width $c_{b_\theta+1} - c_{b_\theta}$, which tends to 0 as $K \rightarrow \infty$; in the continuous limit eligibility coincides with $\tilde{\theta}_i \leq \theta$, so no eligible patient auto-declines. $\square$

Proposition 5 is the engine of the coarsening results in §10. Coarse buckets reintroduce exactly the wasted, cascade-feeding declines that the eligibility filter was meant to remove, and the damage scales with δ .

4.5 Hybrid: threshold + batching

The hybrid combines the eligibility filter of threshold with the within-batch freezing of batching. The eligible set $\mathcal{E}_\theta(\mathcal{Q})$ is partitioned into batches of size m , and within each batch the declines counter is frozen as in Algorithm 2. Formally it is Algorithm 3 with the per-offer update replaced by the batched update of Algorithm 2 applied to $\mathcal{E}_\theta(\mathcal{Q})$ rather than to $\mathcal{Q}$. We evaluate $K=2+m=5$ and $K=10+m=5$ in §10.

5 Simulation and Metrics

Simulation Environment. We simulate a stream of $N = 1500$ organs. The initial queue contains 150 patients with thresholds drawn from (3). After each organ is drawn, $A_k \sim \text{Poisson}(\rho)$ new patients arrive ($\rho = 1.6$). The first 300 organs are warm-up, discarded from analysis so the queue reaches a stationary regime, and the remaining 1200 organs form the test set. We average over 15 independent replications since our trials depend on some randomness. Estimated standard errors on discard rates are 0.001–0.004, so all reported differences exceed Monte Carlo noise by an order of magnitude.

Outcome metrics. Let $\mathcal{K}$ be the set of test-set organs and, for mechanism M , let $\text{alloc}_M \subseteq \mathcal{K}$ be those allocated and n_θ^M the queue position of the recipient of θ . Define

$$D(M) = \frac{|\mathcal{K}| - |\text{alloc}_M|}{|\mathcal{K}|} \quad (\text{discard rate}), \quad (12)$$

$$P(M) = \frac{1}{|\text{alloc}_M|} \sum_{\theta \in \text{alloc}_M} n_\theta^M \quad (\text{mean placement}), \quad (13)$$

$$\text{transplant rate} = \frac{\#\{\text{patients ever allocated}\}}{\#\{\text{patients}\}}, \quad (14)$$

and, for an allocated organ, the recipient **match quality** (welfare surplus) $w = \theta - \tilde{\theta}_{i^*} \geq 0$, with per-organ welfare $\max(\theta - \tilde{\theta}_{i^*}, 0)$ (zero if discarded). Mean wait is the number of organs that arrive between a recipient’s queue entry and their allocation.

Random-number discipline. For the tables below, mechanisms generally run on the same calibrated environment with mechanism-specific acceptance-draw streams. However, for the fairness analysis (§7, §12) we use **common random numbers (CRN)**. This method uses identical organ arrivals, patient arrivals, thresholds, and acceptance draws across mechanisms, enabling patient-by-patient outcome comparison and Pareto statements.

6 Results: Discard Rates, Placements, Transplant Rates

Tables 1–3 report the aggregate metrics across the δ sweep. The two primary axes are discard rate (lower is better) and mean placement (lower means the organ went to a higher-priority patient).

We see four immediate patterns:

Table 1: Aggregate metrics at $\delta = 0$ (no observational learning).

Mechanism	Discard rate	Mean placement	Transplant rate
Omniscient	2.3%	22.2	47.4%
Sequential	9.4%	2.1	41.4%
Batching $m = 5$	5.5%	4.6	44.9%
Threshold $K = 2$ (OPTN)	5.6%	4.6	44.7%
Threshold $K = 5$	5.8%	3.7	44.5%
Threshold $K = 10$	3.8%	7.4	46.3%
Threshold continuous ($K = \infty$)	2.0%	21.7	47.7%

Table 2: Aggregate metrics at $\delta = 0.3$ (moderate observational learning).

Mechanism	Discard rate	Mean placement	Transplant rate
Omniscient	2.3%	22.2	47.4%
Sequential	10.8%	2.1	40.2%
Batching $m = 5$	8.4%	2.6	42.1%
Threshold $K = 2$ (OPTN)	6.5%	4.8	44.0%
Threshold $K = 5$	6.0%	3.9	44.5%
Threshold $K = 10$	3.8%	7.8	46.1%
Threshold continuous ($K = \infty$)	2.2%	21.7	47.5%

(1) **Continuous threshold tracks first-best at every δ .** Its discard rate sits 0.3–0.5 percentage points above omniscient (the irreducible 10% logistic baseline), but the cascading declines effect is essentially neutralized. From $\delta = 0$ to $\delta = 1.0$, its discard rises only 2.0% $\rightarrow$ 2.6%, a +0.6pp sweep (Table 4).

(2) **Batching beats sequential at all δ ,** consistent with Proposition 3. Batching saves ~ 4 pp at $\delta = 0$ and ~ 5 pp at $\delta = 1.0$.

(3) **Coarse thresholds behave non-uniformly in δ .** At low δ the coarse variants ($K = 2, 5$) perform comparably to batching. As δ grows, finer- K thresholds pull ahead while $K = 2$ does not. At $\delta = 1.0$, $K = 2$ (15.1%) is statistically tied with batching (14.4%), whereas $K = 10$ (6.3%) is far better. This demonstrates the within-bucket-cascade mechanism of Proposition 5.

(4) **Mean placement reveals a depth–discard tradeoff.** The mechanisms with the lowest discards (omniscient at placement 22.2, continuous threshold at 19.6) reach deep into the queue, while the shallow mechanisms (sequential at 1.5) discard the most. Whether this deep reach harms front-of-queue patients is a fairness question we resolve in §7. (The answer is no, it does not harm the patients with priority.)

Table 4 makes the central comparative static explicit. We see that **self-selection insulates a mechanism against the cascade**. The continuous threshold moves +0.6pp over the full sweep while sequential moves +10.2pp. Additionally, granularity monotonically shrinks the sweep sensitivity ($K=10$: +2.5pp; $K=5$: +4.9pp), with the binary $K=2$ the exception discussed in §10.

6.1 Patient Wait Times

For a recipient, the *wait* is the number of organs that arrived between joining the queue and being matched. Table 5 reports mean wait and transplant rate at $\delta = 1.0$.

Table 3: Aggregate metrics at $\delta = 1.0$ (strong observational learning).

Mechanism	Discard rate	Mean placement	Transplant rate	Mean wait
Omniscient	2.3%	22.2	47.4%	505
Sequential	19.6%	1.5	32.7%	658
Batching $m = 5$	14.4%	1.7	37.2%	613
Threshold $K = 2$ (OPTN)	15.1%	3.8	36.6%	621
Threshold $K = 5$	10.7%	2.7	40.4%	581
Threshold $K = 10$	6.3%	6.9	44.2%	—
Threshold continuous ($K = \infty$)	2.6%	19.6	47.2%	508

Table 4: Discard rate across the δ sweep, with the change from $\delta = 0$ to $\delta = 1.0$ (Δ over sweep).

Mechanism	$\delta = 0$	$\delta = 0.3$	$\delta = 1.0$	Δ over sweep
Omniscient	2.3%	2.3%	2.3%	0.0 pp
Sequential	9.4%	10.8%	19.6%	+10.2 pp
Batching $m = 5$	5.5%	8.4%	14.4%	+8.9 pp
Threshold $K = 2$ (OPTN)	5.6%	6.5%	15.1%	+9.5 pp
Threshold $K = 5$	5.8%	6.0%	10.7%	+4.9 pp
Threshold $K = 10$	3.8%	3.8%	6.3%	+2.5 pp
Threshold continuous ($K = \infty$)	2.0%	2.2%	2.6%	+0.6 pp

Better mechanisms deliver *both* shorter waits and higher transplant rates; continuous threshold matches omniscient on both. The collapse under sequential at strong cascading declines effect is severe: only 33% of patients ever receive an organ within the horizon, and those who do wait 658 organs on average. Here, we see that preventing discards translates directly into shorter waits and more transplants, with no efficiency–equity tension at the aggregate level.

7 Is Anyone Worse Off: Patient-Level Pareto Analysis

The aggregate numbers could in principle mask a distributional concern: perhaps threshold mechanisms cut discards by diverting organs away from the priority-1 patients who would have received them under batching or sequential. We test this directly with common random numbers (CRN), tracking each individual patient’s outcome under each mechanism on the same realized environment.

Definition 3 (Patient-level Pareto comparison). Fix two mechanisms A, B and a realized environment. Partition patients into four cells by whether they are transplanted under each: *both*, *only A*, *only B*, *neither*. A **strictly Pareto-dominates** B on this environment iff $|\text{only } A| > 0$ and $|\text{only } B| = 0$: some patient is helped by switching to A and none is harmed. If both “only” cells are nonempty the mechanisms are *non-comparable*.

Summing across the 8 CRN replications as in Figure 3, we find that:

- $\delta = 0$. Threshold continuous strictly dominates batching (341 better, 0 worse) and sequential (899 better, 0 worse). Threshold $K=2$ vs. batching is non-comparable (batching helps 82 that $K=2$ does not; $K=2$ helps 2 that batching does not).

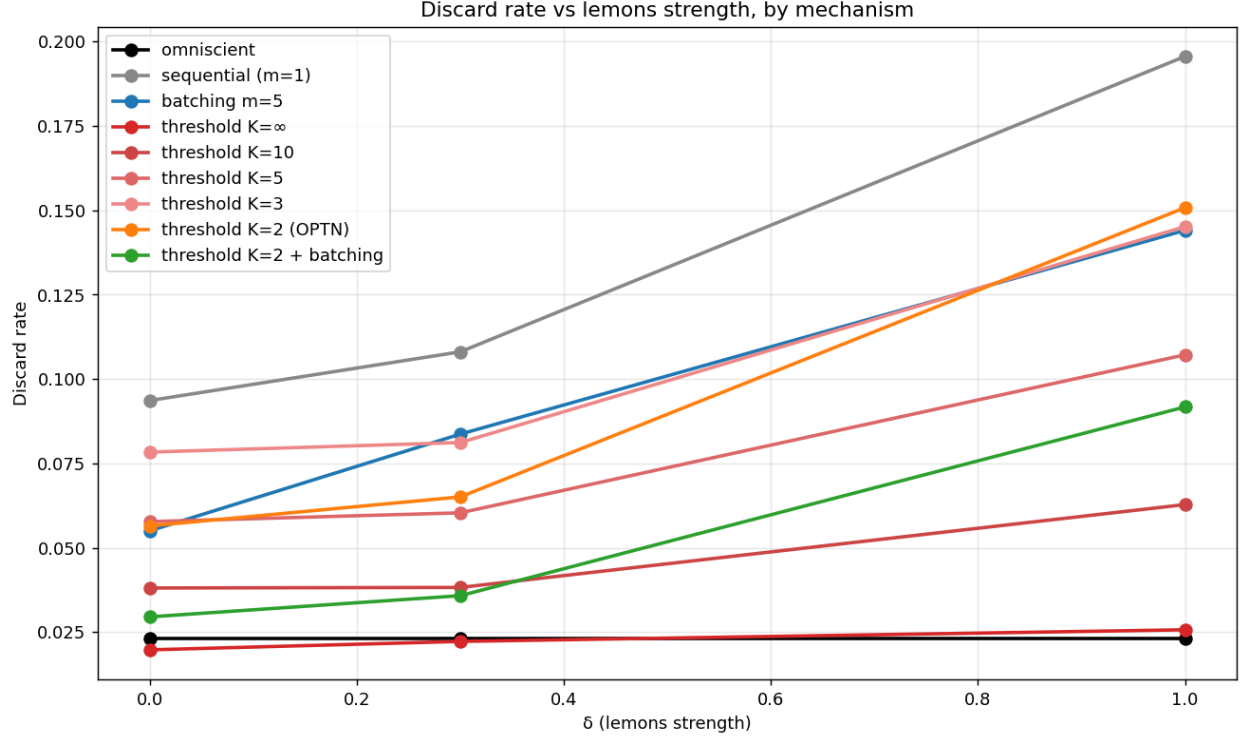


Figure 1: **Discard rate as a function of cascading declines strength δ , by mechanism.** Threshold continuous (dark red) and omniscient (black, dotted, largely hidden behind it) are flat near 2%. Sequential (gray) climbs to nearly 20% at $\delta = 1.0$; batching ($m = 5$, blue) rises more slowly. The coarse thresholds ($K = 10, 5, 3, 2$) form an intermediate band whose slopes steepen as K falls. The hybrid threshold $K=2$ +batching (green) lies below both pure batching and pure $K=2$ throughout.

- $\delta = 0.3$. Threshold $K=2$ strictly dominates batching (238 better, 0 worse) and sequential (559 better, 0 worse). Threshold continuous strictly dominates batching (748 better, 0 worse).
- $\delta = 1.0$. Threshold continuous strictly dominates batching (1410 better, 0 worse) and sequential (2026 better, 0 worse). $K=2$ vs. batching is again non-comparable ($K=2$ helps 86, batching helps 21).

Proposition 6 (Strict Pareto dominance of continuous threshold). *On every CRN environment tested with $\delta > 0$, threshold continuous strictly Pareto-dominates batching, sequential, and every coarse threshold variant: every patient transplanted under any of those mechanisms is also transplanted under threshold continuous, and strictly more patients are transplanted in total.*

Notably, **the extra placement depth of threshold mechanisms is not redistribution but recovery.** The organs that continuous threshold places deep in the queue are organs that batching and sequential *discarded*. The front-of-queue patients who received their (better) organs under batching receive the same organs under threshold, so no priority-1 patient is bumped down. This is a far stronger claim than a lower average discard rate, and it holds patient by patient under identical randomness.

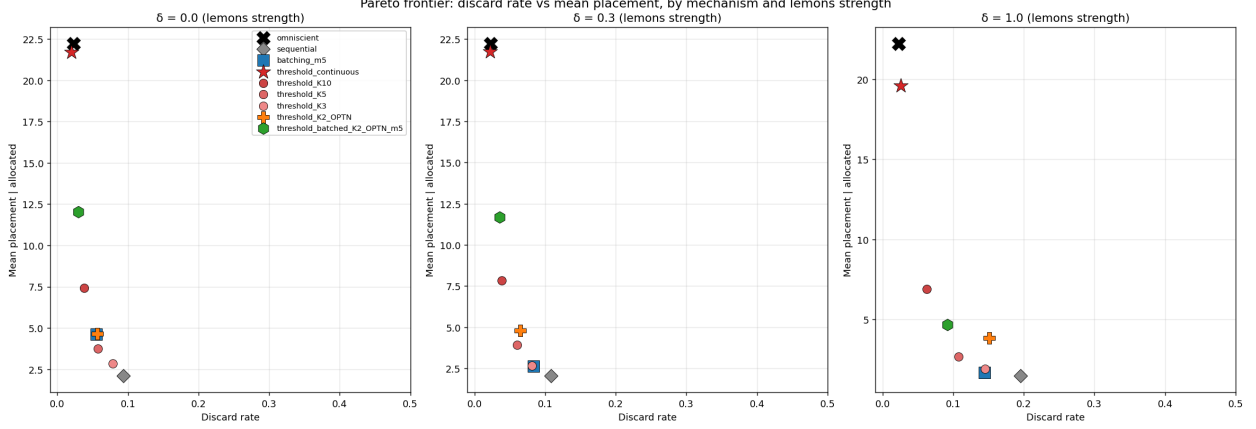


Figure 2: **Pareto frontier in (discard rate, mean placement) space**, separately for $\delta = 0, 0.3, 1.0$. Lower-left is better. Omniscient (black X) and continuous threshold (red star) sit at high placement but very low discard. Sequential (gray diamond) and batching (blue square) sit at low placement but high discard, moving rightward as δ grows. The coarse thresholds interpolate. As δ rises the frontier shifts right for mechanisms without self-selection while threshold continuous stays put.

Table 5: Mean wait (in organs) and transplant rate at $\delta = 1.0$.

Mechanism	Mean wait	Transplant rate
Omniscient	505	47.4%
Sequential	658	32.7%
Batching $m = 5$	613	37.2%
Threshold $K = 2$ (OPTN)	621	36.6%
Threshold $K = 5$	581	40.4%
Threshold continuous ($K = \infty$)	508	47.2%

8 Mechanism Ordering Performance Analysis

At moderate-to-strong learning ($\delta \geq 0.3$) the mechanisms order from best-to-worst as

$$\text{Threshold } K=\infty \succ K=10 \succ K=5 \succ \text{Batching} \gtrsim K=2 \text{ (OPTN)} \succ \text{Sequential},$$

while at $\delta = 0$ the non-sequential mechanisms are comparable. The ordering follows from a single accounting identity. Every decline by a non-accepting patient costs the system twice, since it consumes one period of the H -period budget, and it raises the previous declines count k for the next patient by one. Mechanisms that minimize *unnecessary* declines (offers to patients whose threshold the organ does not meet) win on both margins simultaneously.

The cleanest summary is the **gap to first-best**, $D(M) - D(M_{\text{Omn}})$, which by Proposition 1 is pure information-friction waste. At $\delta = 1.0$ (omniscient discard = 2.3%):

The decomposition shows that the bulk of the inefficiency in current allocation comes from offering to non-acceptors. Sequential offers to everyone in line: a patient with $\tilde{\theta}_i = 0.80$ offered $\theta = 0.10$ declines, costing a period and a unit of cascade. Batching offers to five-at-a-time but still wastes offers on threshold violators (now five at a time), buying back some loss through information-freezing. Threshold mechanisms route the organ directly to declared acceptors; with continuous

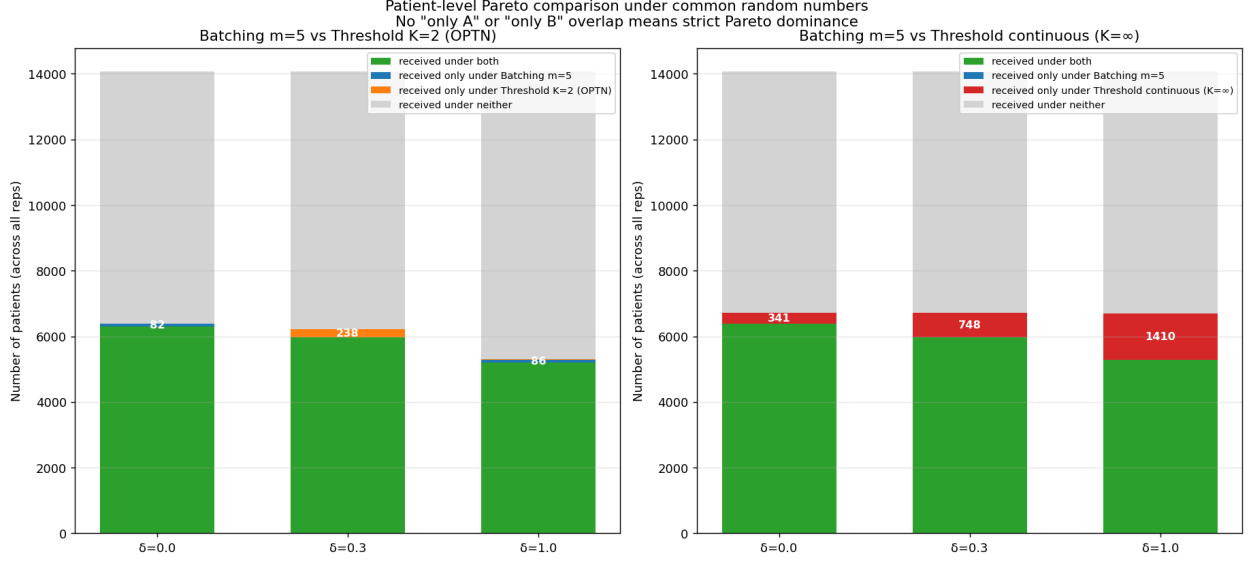


Figure 3: **Patient-level Pareto comparison under common random numbers.** Each stacked bar splits patient outcomes for a mechanism pair at one δ : green = received under both, blue/orange/red = received under only one, gray = neither. A zero “only-B” segment means A strictly dominates. *Left*: batching vs. $K=2$ (OPTN). *Right*: batching vs. threshold continuous, which helps strictly more patients at every δ (341, 748, 1410) with zero worse off.

Table 6: Gap to first-best (excess discard over omniscient) at $\delta = 1.0$.

Mechanism	Gap to omniscient
Threshold continuous	0.3 pp
Threshold $K = 10$	4.0 pp
Threshold $K = 5$	8.4 pp
Batching $m = 5$	12.1 pp
Threshold $K = 2$ (OPTN)	12.8 pp
Sequential	17.3 pp

declarations essentially no offer is wasted, and even moderate granularity ($K=10$) recovers most of the first-best gain. **Once patients can opt in or out even moderately finely, we capture a large share of the omniscient algorithm’s gains.**

9 The Batch-Size Margin

Before turning to declaration granularity, we examine the batching margin. We examine how performance varies with the batch size m , holding the mechanism family fixed. Sequential is the $m = 1$ special case. Table 7 and Figure 4 sweep $m \in \{1, 2, 3, 5, 8, 12\}$ at each δ .

We record three main observations of our results. **(1) Discard falls monotonically in m** at every δ . Wider simultaneous offers always help by enlarging the reachable pool ($H \cdot m$ candidates) and freezing more of the cascade within each batch. **(2) The marginal value of batching rises with δ .** Going from $m=1$ to $m=12$ saves 6.4pp at $\delta = 0$ but 10.3pp at $\delta = 1.0$, because a larger frozen batch suppresses more cascade signal exactly when the cascade is most damaging.

Table 7: Batch-size sweep at $\delta = 1.0$. $m = 1$ is sequential. Larger m reaches more candidates and freezes more of the cascade within each batch.

Batch size m	Discard	Transplant rate	Mean placement
1 (sequential)	20.8%	46.7%	1.5
2	18.8%	47.9%	1.5
3	17.3%	48.7%	1.6
5	15.1%	50.1%	1.7
8	12.4%	51.7%	1.9
12	10.5%	52.8%	2.2

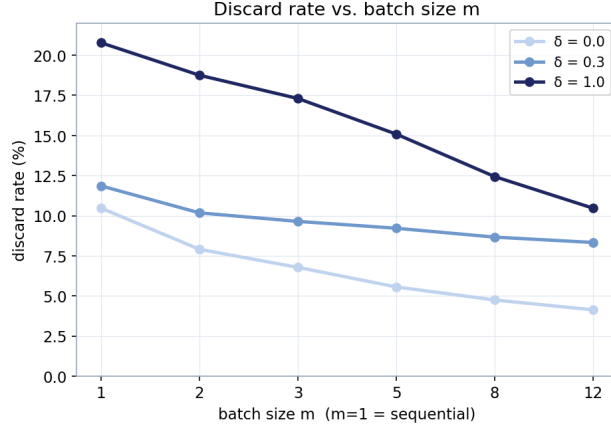


Figure 4: **Discard rate versus batch size m** , one line per δ (presentation slide 7). Discard falls monotonically in m at every δ ; the curves are steepest at $\delta = 1.0$ (top, navy), where freezing more of the cascade within a batch is most valuable.

(3) **However, batching cannot fully escape the cascade.** Since k still updates *across* batches at rate m per period (Algorithm 2), even $m = 12$ leaves a 10.5% discard at $\delta = 1.0$, far above threshold $K=10$'s 6.3%. Additionally, deeper batches mildly raise mean placement. The contrast with the threshold family is instructive as to the mechanisms which lead to improved performance in both. Batching attacks the cascade by *hiding* declines, whereas the eligibility filter attacks it by *preventing* the declines from being generated at all. The latter is the more powerful lever, which is why the hybrid of §10 dominates either alone.

10 Declaration Granularity and the Hybrid Mechanism

How coarse can the declaration be while still capturing the gains? We sweep $K \in \{2, 3, 5, 10, \infty\}$, focusing on how coarsening interacts with δ .

The pattern (Figure 5) shows that **discard is decreasing in K , with one placement-driven exception, and the cost of coarsening grows with δ** . Going from $K = \infty$ to $K = 2$ costs

$$D(K=2) - D(K=\infty) = \begin{cases} 3.6 \text{ pp}, & \delta = 0, \\ 4.3 \text{ pp}, & \delta = 0.3, \\ 12.5 \text{ pp}, & \delta = 1.0. \end{cases} \quad (15)$$

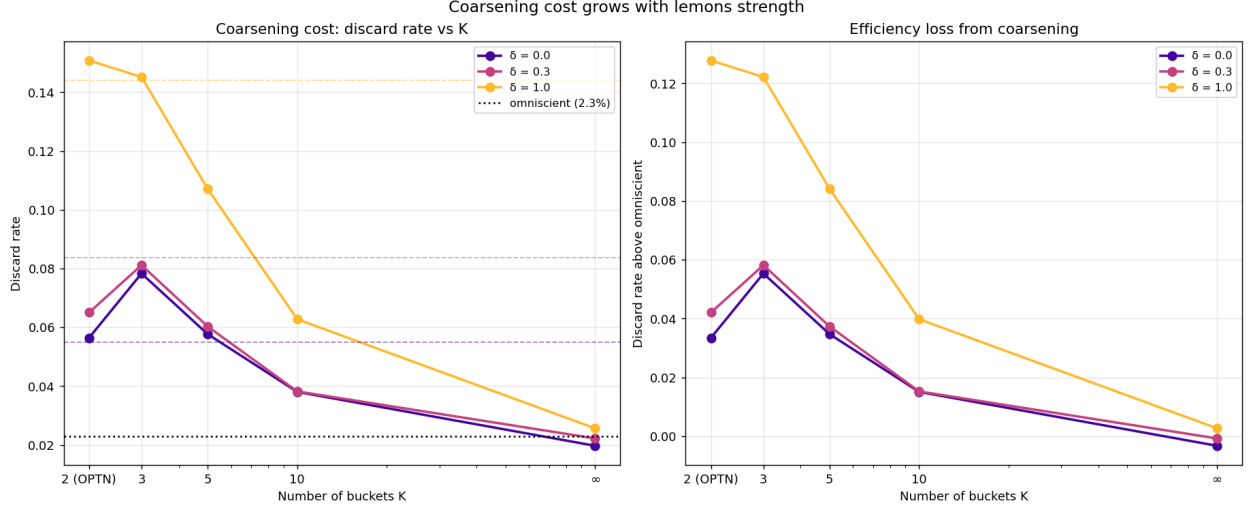


Figure 5: **Coarsening effect on threshold mechanisms.** *Left:* discard rate versus K (log-spaced), one line per δ . *Right:* efficiency loss (discard above omniscient) versus K . At $\delta = 0$ the loss is small at every K . At $\delta = 1.0$ it grows sharply as K falls. Note the non-monotonicity at $K = 3$ since uniform cutoffs at $1/3, 2/3$ do not match the right-skewed threshold distribution.

The mechanism is what is described in Proposition 5. We see that coarse buckets admit within-bucket auto-declines, each of which both wastes an offer. Furthermore, when δ is large, this triggers a compounding cascade. When δ is small a within-bucket decline is merely a lost offer. When δ is large it is an amplified disaster. Even the modest move from $K=2$ to $K=5$ recovers meaningful efficiency at $\delta = 1.0$ (10.7% vs. 15.1%).

Remark 2 (Cutoff placement dominates bucket count: the $K = 3$ anomaly). $K = 3$ underperforms $K = 2$ (OPTN) at every δ (e.g., 7.8% vs. 5.6% at $\delta = 0$). The $K = 3$ cutoffs at $\theta = 1/3, 2/3$ sit in the thin right tail of the Beta(2, 5) threshold distribution (mean 0.286), producing badly unbalanced buckets, whereas the OPTN binary cutoff at $\theta = 0.15$ falls in the heart of the distribution and yields an informative split. This demonstrates that **the placement of cutoffs matters more than the number of splits**. A well-chosen 2-bucket question beats a poorly-chosen 3-bucket one. A direct cutoff sweep over $\theta \in \{0.15, 0.20, 0.30, 0.40, 0.50\}$ confirms that the OPTN value 0.15 minimizes discard among binary cuts at every δ , which is a nontrivial validation of existing policy. This is why the looser phrasing “more granularity always helps” should be qualified: granularity helps conditional on sensible cutoff placement. This implies that a possible direction of further research is to optimize the split location over the full range of possible θ values rather than choosing uniform splits.

10.1 The Hybrid Model: Threshold $\times$ Batching

We established that the eligibility filter and batch-freezing attack different sources of loss (Proposition 5 vs. the across-batch freezing of Algorithm 2), so combining them should compound. Table 8 reports the hybrid against its pure components at $\delta = 1.0$.

The hybrid strictly improves on both components at each granularity. At $K = 2$ the hybrid cuts discard to 9.8%, well below both pure batching (15.1%) and pure $K = 2$ (15.4%), while raising the transplant rate to 53.2%. At $K = 10$ it reaches 3.4%, approaching continuous threshold’s 2.6%. Under CRN the hybrid helps strictly more patients than batching does (637/0 at $\delta = 1.0$) and than pure threshold (676/0) does, with no patients worse off in either regime. The cost of this lower

Table 8: Hybrid threshold $\times$ batching versus pure components, $\delta = 1.0$.

Mechanism ($\delta = 1.0$)	Discard	Transplant rate	Mean placement
Batching $m = 5$	15.1%	50.1%	1.7
Threshold $K = 2$	15.4%	49.9%	4.1
Hybrid $K = 2 + m = 5$	9.8%	53.2%	5.4
Threshold $K = 10$	6.6%	55.2%	7.9
Hybrid $K = 10 + m = 5$	3.4%	57.0%	18.6

discard rate is depth, as mean placement rises (to 18.6 at $K=10+m=5$), the same recovery-not-redistribution depth documented in §7 and §11. We reaffirm that **batching and threshold are complements, not substitutes**. Self-selection prevents wasted offers, and batching freezes the residual within-bucket cascade that self-selection alone leaves behind.

11 Marginal Kidneys

The original motivation for this study is the marginal-organ problem. About 45% of high-KDPI organs (defined as $KDPI \geq 85$, our $\theta \leq 0.15$) are discarded in current U.S. practice (Mohan et al., 2018), versus under 5% of premium organs. Because the aggregate discard rate is dominated by these marginal organs, they are where mechanism choice has its largest effect. For this part of the analysis, we restrict attention to marginal organs ($\theta \leq 0.15$, about 15% of arrivals) and ask three questions: how often each mechanism allocates them, where they go in the queue, and how well-matched the recipients are.

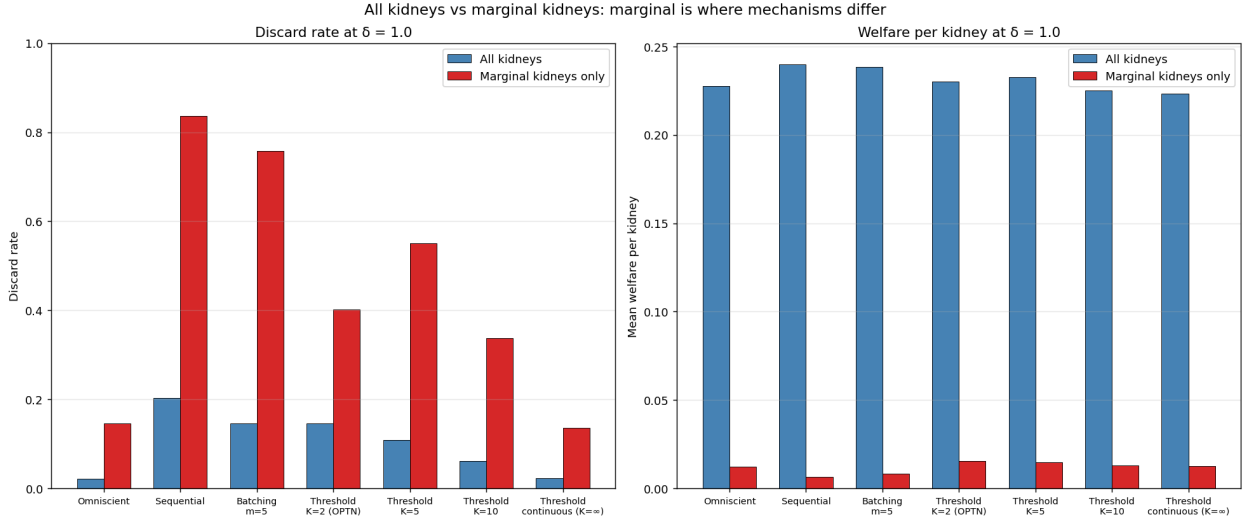


Figure 6: **All organs vs. marginal organs** at $\delta = 1.0$. *Left*: discard rate. The marginal (red) bars tower over the all-organ (blue) bars under sequential and batching but are nearly equal under threshold continuous. *Right*: welfare per organ. All-organ welfare is dominated by premium organs (similar across mechanisms), while marginal-organ welfare separates the mechanisms cleanly.

11.1 Marginal-Organ Discard

The aggregate pattern is amplified for marginal organs. At $\delta = 1.0$:

Table 9: Marginal-organ discard rate at $\delta = 1.0$ ($\theta \leq 0.15$).

Mechanism	Marginal-organ discard
Omniscient	14.5%
Threshold continuous ($K = \infty$)	13.7%
Threshold $K = 10$	33.7%
Threshold $K = 2$ (OPTN)	40.1%
Threshold $K = 5$	55.1%
Batching $m = 5$	75.8%
Sequential	83.6%

Compared with the population-wide rates at $\delta = 1.0$ (batching 14.4%, $K=2$ 15.1%, continuous 2.6%), the mechanisms differ *far* more on marginal organs. Mechanism choice moves the marginal-organ discard rate by roughly 70 percentage points (sequential 83.6% vs. continuous 13.7%), against only about 17 points population-wide. **This shows that marginal organs are where the mechanism makes the most difference.**

Omniscient itself discards $\approx 14\%$ of marginal organs. These are organs for which no waiting patient has a low enough threshold, genuinely unallocatable given the population. We see that continuous threshold essentially attains this floor, though it is unattainable in real life. Coarser thresholds still suffer within-bucket cascades (Proposition 5). However, they perform better than the other methods, where batching and sequential fail catastrophically because they spend the budget offering marginal organs to high-threshold patients who will always decline.

Remark 3 (A second placement-driven non-monotonicity). *On marginal organs, $K = 2$ (OPTN) beats $K = 5$ (40.1% vs. 55.1%), reversing their aggregate order (15.1% vs. 10.7%). The reason is again placement: the $K = 2$ cutoff at $\theta = 0.15$ exactly defines the marginal boundary, so the marginal bucket $[0, 0.15)$ contains only patients who genuinely want marginal organs. The $K = 5$ bucket $[0, 0.2)$ additionally contains patients with $\tilde{\theta}_i \in (0.15, 0.20)$ who decline the most marginal organs, generating within-bucket cascades precisely on the organs that can least afford them. The OPTN cutoff, by accident or design, is placed where it does the most good for the organs that need the most help. This is the precise statement behind the presentation’s slide-15 caveat that granularity helps “except for $K = 2$ at the OPTN level”. We see that $K = 2$ is a favorable exception on marginal organs.*

11.2 Marginal Organ Placement and Quality

For marginal organs (low θ), recipients must have low thresholds. At every δ the mean recipient threshold for marginal organs is $\tilde{\theta}_{i*} \approx 0.07$ across all mechanisms, which is well below the population mean 0.286. This means that marginal organs uniformly reach patients who explicitly want them.

Match quality is positive everywhere, so **no patient receives an organ they did not want**. This is due to the gate’s explicit *ex post* rationality property. However, we see that the ordering is the *reverse* of the discard ordering. This is because sequential and batching attain high *per-match* surplus precisely because they fail so often. These mechanisms place a marginal organ only when they happen to reach an extremely-low-threshold patient, leaving a large gap $\theta - \tilde{\theta}_{i*}$. Threshold

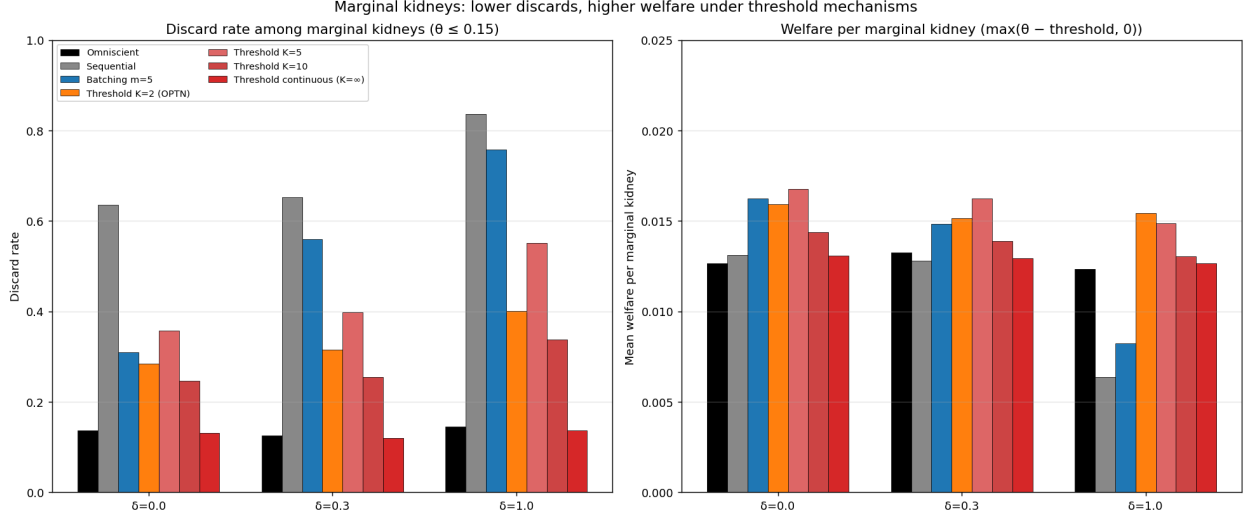


Figure 7: **Marginal-organ discard rate and welfare per marginal organ**, by mechanism and δ . *Left*: discard; sequential discards 64–84% across δ , batching 31–76%, $K=2$ 28–40%, with continuous and omniscient at the 13–15% floor. *Right*: welfare per marginal organ $\max(\theta - \tilde{\theta}_{i^*}, 0)$. Threshold mechanisms produce 2–3 $\times$ the total welfare of sequential or batching, despite reaching further into the queue.

Table 10: Mean match quality $\theta - \tilde{\theta}_{i^*}$ for marginal organs at $\delta = 1.0$.

Mechanism	Mean match quality (marginal organs)
Sequential	+0.039
Batching $m = 5$	+0.034
Threshold $K = 2$ (OPTN)	+0.026
Threshold continuous ($K = \infty$)	+0.015
Omniscient	+0.014

mechanisms place far more marginal organs (higher allocation rate, lower discard) by also serving patients whose threshold is only slightly below the organ quality, which lowers *mean* surplus while raising *total* welfare. This is an allocation-rate vs. surplus-per-match decomposition, not a coercion effect.

11.3 Allocation Depth by Organ Quality

Figure 9 resolves the fairness story geometrically. For premium organs ($\theta > 0.7$) every mechanism allocates at position 1–2. This makes sense, as the front of the queue accepts a good organ immediately, so no mechanism reaches deep. The mechanisms really diverge only for marginal organs ($\theta < 0.15$), where batching reaches positions ~ 10 –25 and threshold continuous reaches 100+. **The deep reach of threshold mechanisms is exclusively a marginal-organ phenomenon.** Combined with Proposition 6, this means the deep recipients are not displacing front-of-queue patients. The patients those deep organs “skip” are patients who would have declined anyway (their declared bucket excludes the organ, or their threshold does). The organs reaching position 100+ are organs that would otherwise have been discarded, which are exactly the organs we wanted to allocate when we motivated this problem.

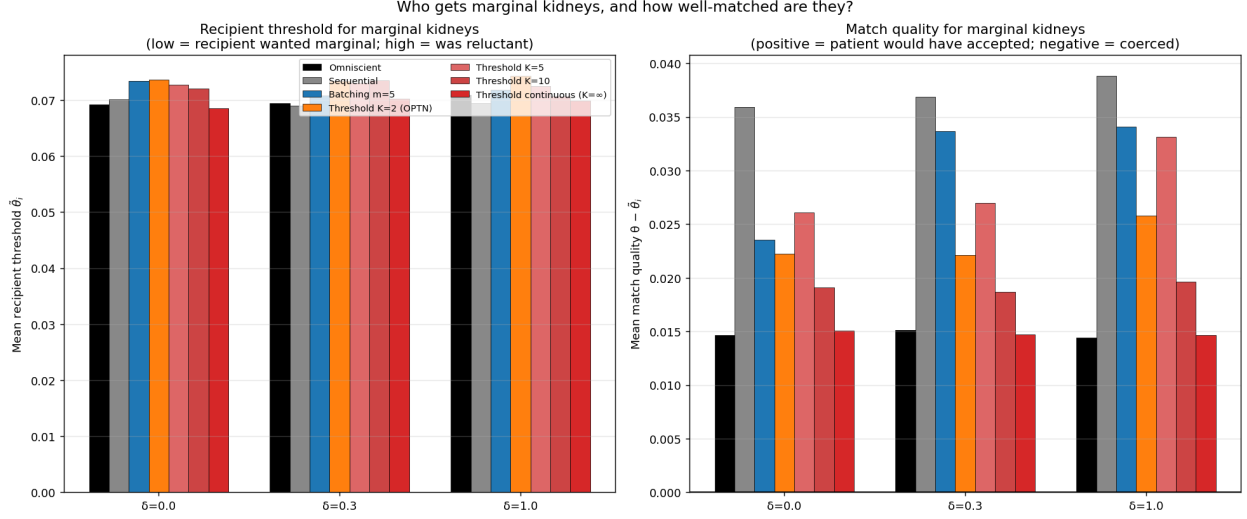


Figure 8: **Recipient threshold (left) and match quality $\theta - \tilde{\theta}_{i^*}$ (right) for marginal organs.** Match quality is positive in every mechanism, so no recipient is coerced. This confirms the *ex post* rationality guaranteed by the threshold gate. Sequential and batching show *higher* surplus per allocation because they only manage to place marginal organs with the rare very-low-threshold patients (everyone in between declined). In contrast, threshold mechanisms place across a wider band of low-threshold patients, including those whose threshold sits just below the organ quality.

11.4 Coarsening on Marginal Organs, and Welfare per Marginal Organ

Combining discard and match quality gives welfare per marginal organ, $\max(\theta - \tilde{\theta}_{i^*}, 0)$ averaged over marginal organs. At $\delta = 1.0$, we see that Table 11 shows that welfare is highest for low- K threshold mechanisms.

Table 11: Welfare per marginal organ at $\delta = 1.0$.

Mechanism	Welfare per marginal organ
Threshold $K = 2$ (OPTN)	0.015
Threshold $K = 5$	0.015
Threshold $K = 10$	0.013
Threshold continuous ($K = \infty$)	0.013
Omniscient	0.012
Batching $m = 5$	0.008
Sequential	0.006

Threshold mechanisms produce roughly $2\times$ the welfare per marginal organ of batching and $2\text{--}3\times$ that of sequential. The coarse variants ($K=2, 5$) edge out continuous and omniscient on this *per-organ* metric for the surplus-vs-rate reason above. Specifically, continuous allocates a far higher fraction of marginal organs (87% vs. $K=5$'s 45%) but matches each to a patient whose threshold ($\tilde{\theta}_{i^*} \approx 0.07$) is close to the organ quality, leaving less per-match surplus.

Remark 4 (Two objectives, two winners). *Omniscient and continuous threshold maximize the marginal-organ allocation rate (and minimize discards). On the other hand, coarse threshold mechanisms maximize per-match surplus. Which is preferable will depend on the planner's objective, but*

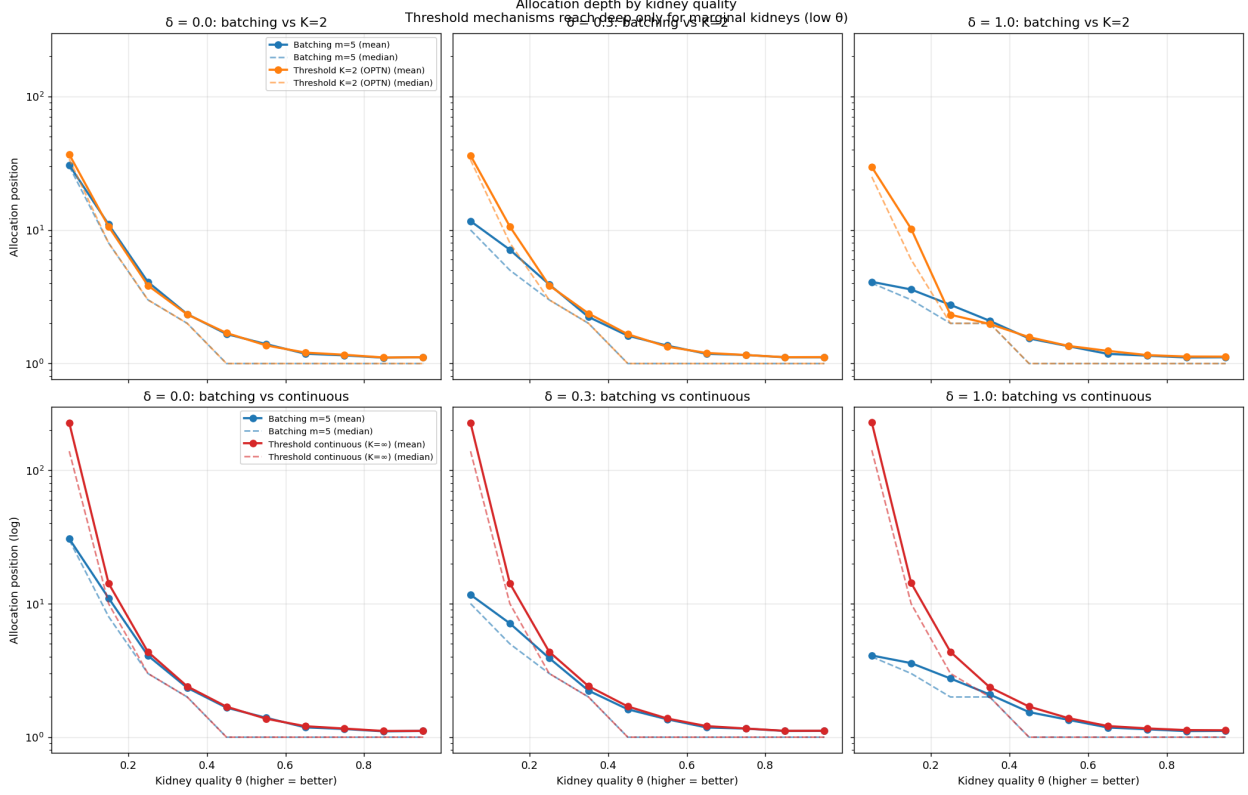


Figure 9: **Allocation depth as a function of organ quality**, by mechanism and δ (presentation slide 17). *Top*: batching vs. $K=2$. *Bottom*: batching vs. threshold continuous (log y -axis). Premium organs ($\theta > 0.7$) are placed at position 1–2 under every mechanism. The mechanisms diverge only on marginal organs, where threshold continuous walks to position 100+ to find a declared acceptor.

the differences in the per-match metric are observably small (0.012–0.015) and are dominated by the order-of-magnitude differences in allocation rate. Under a rough aggregate welfare objective (total surplus = allocation rate $\times$ surplus per match) the high-allocation mechanisms win.

12 Granularity Versus Greedy Mechanisms: CRN Analysis

Proposition 6 concerns the (unattainable) continuous benchmark. For policy we care about *implementable* declarations, so we run the patient-level CRN comparison of the two leading deployable threshold variants— $K=2$ (today’s OPTN policy) and $K=10$ (a realistic finer opt-in)—against the two greedy mechanisms. Table 12 reports, for each pair and each δ , the count of patients helped only by the threshold variant versus only by the greedy mechanism, summed over 8 CRN replications.

$K = 10$ **strictly Pareto-dominates both greedy mechanisms at every δ** . The “only-greedy” count is zero in all six $K=10$ cells, while the “only- $K=10$ ” count rises with δ (to 1,724 vs. sequential and 1,029 vs. batching at $\delta = 1.0$). The finer opt-in never makes a patient worse off and helps many more, increasingly so as the cascade strengthens.

$K = 2$ **dominates sequential everywhere but is non-comparable with batching** at $\delta = 0$ (1/113) and $\delta = 1.0$ (20/65), dominating batching only at the intermediate $\delta = 0.3$ (255/0). This is the same non-monotonicity flagged in §7. We see that today’s binary opt-in does *not* uniformly beat batching at the patient level. **Instead, it is the move to a finer opt-in ($K=10$) that converts**

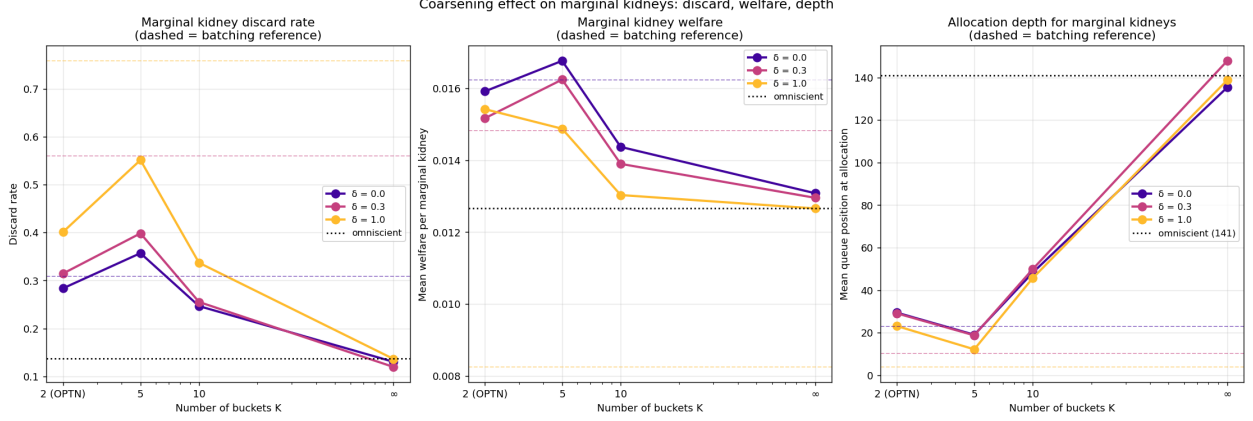


Figure 10: **Coarsening effect restricted to marginal organs:** discard (left), welfare per organ (middle), and mean allocation position (right) versus K . $K = 2$ achieves lower marginal-organ discard than $K = 5$ at every δ (Remark 3). Welfare per marginal organ peaks at coarse K . Mean position rises sharply with K , reaching 130–150 at $K = \infty$ for the most marginal organs.

Table 12: Patient-level CRN dominance: (patients better only under threshold / better only under greedy), summed over 8 replications. A zero second entry means the threshold variant strictly Pareto-dominates.

Pairing (threshold / greedy)	$\delta = 0$	$\delta = 0.3$	$\delta = 1.0$
$K = 2$ (OPTN) vs. Sequential	479/0	564/0	650/0
$K = 2$ (OPTN) vs. Batching	1/113	255/0	20/65
$K = 10$ vs. Sequential	753/0	910/0	1,724/0
$K = 10$ vs. Batching	162/0	601/0	1,029/0

the tie into clean dominance. This means that the policy recommendation is therefore not “adopt threshold” in the abstract but “refine the existing opt-in question,” which both dominates the greedy alternatives and is incrementally implementable on top of current OPTN infrastructure.

13 Additional Notes on the Behavioral Equation

We also want to note that the acceptance rule (4) is a synthesis of three established literatures. No single paper uses this exact functional form, but each component is standard. Here, we will state the lineage of each component.

The threshold gate ($\tilde{\theta}_i > \theta \Rightarrow \text{reject}$). A reservation quality from optimal-stopping / search theory. The reservation-value structure originates with McCall (1970) in job search. Its kidney-specific form, specifically a candidate choosing whether to accept an organ now or wait for a better one given queue dynamics, is the acceptance-threshold policy of Su and Zenios (2004), after whom our continuous ($K = \infty$) variant is named.

The logistic layer $\sigma(\alpha + \beta(n - 1) - \delta k)$. A random-utility / binary logit acceptance probability, the workhorse discrete-choice model of McFadden (1974). In the kidney-offer setting specifically, Agarwal et al. (2021) model organ-offer acceptance as a function of organ and patient characteristics.

Our α (baseline) and β (position/desperation) terms are the reduced-form counterpart of that structural object.

The cascade term $-\delta k$. Observational learning. The theory that rational agents observing predecessors’ choices can herd is the informational-cascade framework of Banerjee (1992) and Bikhchandani, Hirshleifer, and Welch (1992). The kidney-specific empirical evidence is Zhang (2010). As shown in (5), the linear-in- k specification is the exact posterior log-odds of a Bayesian patient who treats each prior decline as a conditionally independent, equal-strength signal of a latent organ defect.

Remark 5 (The linear-in- k cascade is a deliberate reduced form). *A fully rational learner would weight declines by their informativeness, which need not be constant. Under herding, later declines are partly imitation and carry less fresh information, suggesting a concave (sublinear) dependence such as δk^γ with $\gamma < 1$ or $\delta \log(1 + k)$. A momentum story suggests convex $\gamma > 1$. We adopt the linear form because it (i) equals i.i.d.-signal Bayesian updating, (ii) nests the no-learning benchmark cleanly at $\delta = 0$, and (iii) keeps a single interpretable intensity parameter. Because the functional form sets the magnitude of the headline effects, we treat the δ -sweep itself as the robustness check: reporting $\delta \in \{0, 0.3, 1.0\}$ brackets the no-learning and strong-learning regimes, and the mechanism ordering (Proposition 6) is invariant across the sweep because self-selection removes the cascade-feeding declines regardless of how k enters the index. Future research may explore non-reduced or alternative forms.*

14 Limitations

Several simplifications bound the interpretation of our results and opens doors for future work.

Identification of the learning channel. The decisive empirical question is whether $\delta > 0$ describes reality. Zhang (2010) reads decline cascades as observational learning ($\delta > 0$). However, Agarwal et al. (2021) read the same data as private organ characteristics ($\delta = 0$ with unobserved heterogeneity). Our results are sharpest under Zhang and reduce to the $\delta = 0$ column under Agarwal. Discriminating between the two requires panel data on offer responses with rich organ covariates, and is the most important empirical extension.

Exogenous, known thresholds. Real patients’ thresholds drift with health and would respond strategically to mechanism changes. The Su–Zenios incentive-compatibility result (Proposition 4) assumes patients know their own $\tilde{\theta}_i$, which the structural choice literature (Agarwal et al., 2025) addresses with explicit choice modeling.

Organ quality. We collapse all medically relevant features into one θ . Real organs are multi-attribute (HLA match, donor age, HCV status, cold-ischemic time), and real opt-ins are multi-dimensional and have far more nuance. Extending threshold declarations to many attributes is straightforward as separate binary opt-ins but more delicate as a joint declaration as incentive-compatibility is no longer guaranteed.

Time-on-list priority only. We abstract from medical urgency, geography, pediatric status, and HLA priority. Some omitted factors correlate with thresholds (very sick patients have both low thresholds and high priority), which would couple the priority and threshold dimensions our model treats as independent.

Quality-invariant δ . We assume the cascade strength is constant across organ quality. If learning operates more strongly on marginal organs (where “why did they decline?” carries more content) a quality-varying δ would *amplify* the threshold advantage on exactly the organs where it already dominates.

Calibrated baseline decline. The $\alpha = \text{logit}(0.9)$ baseline is plausible but not fit to data. Sensitivity is moderate, and shifting the baseline to 80% or 95% changes magnitudes but not the qualitative ordering.

Within these limitations, the qualitative findings are robust. The strict Pareto dominance of continuous threshold under CRN at $\delta > 0$ (Proposition 6) does not depend on calibration choices, and the deployable recommendation (refine the OPTN opt-in granularity) rests on the $K=10$ dominance of Table 12, which notably holds at every δ tested.

15 Conclusion

We compared four mechanisms (omniscient, sequential, batching, and a granularity-indexed threshold family) for allocating perishable kidneys under a unified behavioral model in which patients hold private reservation thresholds and prior declines update beliefs about organ quality. Our main findings are that:

1. **Continuous threshold declaration effectively eliminates discard from observational learning**, attaining first-best discard rates at every δ (Tables 3, 4).
2. **Among deployable rules, a finer opt-in ($K=10$) strictly dominates both greedy mechanisms at every δ , while today’s binary $K=2$ only ties batching** (Table 12). The granularity refinement is what converts the tie into strict dominance.
3. **The cost of coarsening grows with δ , but cutoff placement dominates bucket count** (Eq. (15), Remarks 2, 3). The OPTN cutoff at KDPI=85 is near-optimal among binary cuts and is especially well-placed for marginal organs. Further refining of cutoffs may yield more improvements.
4. **Mechanism choice matters almost entirely on marginal organs** (a $\sim 70\text{pp}$ spread vs. $\sim 17\text{pp}$ population-wide), and the threshold family’s deep reach is confined to marginal organs. Premium organs go to the front under every mechanism (§11).
5. **Batching and threshold are complements.** a hybrid strictly improves on either alone (Table 8).

The current OPTN binary opt-in for high-KDPI organs is, in our framework, the right *kind* of mechanism but coarser than ideal. Moving to a finer opt-in ($K=5$ or $K=10$) would capture substantial additional efficiency, especially if observational learning is as strong as Zhang (2010) estimates. Whether the gains materialize in practice hinges on the learning-versus-private-information question (Agarwal et al., 2021). Further research would require offer-level panel data and remains the central empirical question that could be expanded upon after this work.

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